

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – MenoStart -
Empowering Women to Thrive Through Every Stage of Menopause

Domain of Project – Healthcare

Faculty Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

Problem Statement: AI-Powered Menopause Wellness Agent

The Challenge

Menopause is a natural transition that affects millions of women worldwide, yet it remains underrecognized and poorly supported. Women often experience physical, emotional, and cognitive symptoms during **pre-menopause, perimenopause, menopause, and post-menopause**, making it difficult to understand their current stage and manage their health effectively. Reliable, evidence-based information is scattered across multiple sources, and most existing solutions provide generic advice rather than personalized guidance. Without continuous, individualized support, women may struggle with symptom management, nutrition, mental well-being, fitness, sleep, and workplace productivity.

Objectives

- **Educate & Assess:** Help women understand menopause and identify their current stage (pre-, peri-, menopause, or post-menopause).
- **Personalized Wellness:** Provide tailored recommendations for nutrition, exercise, sleep, mental well-being, and self-care.
- **Continuous Support:** Enable symptom tracking, proactive health insights, and preventive care guidance.
- **Empower Women:** Promote informed decision-making, healthier lifestyles, and improved workplace well-being through trusted, evidence-based AI support.

Proposed Solution -

Proposed Solution: AI-Powered Menopause Wellness Agent

The Objective

Build an **AI-powered Menopause Wellness Agent** that delivers personalized, evidence-based guidance to support women throughout their menopause journey. The solution should include:

- **Menopause Stage Assessment** – Help users identify whether they are in pre-menopause, perimenopause, menopause, or post-menopause based on symptoms, age, menstrual history, and lifestyle.
- **Personalized Wellness Plans** – Recommend daily nutrition, exercise, sleep, stress management, hydration, and self-care routines tailored to individual health conditions, preferences, and goals.
- **Evidence-Based Knowledge RAG Retrieval** – Retrieve and summarize trusted medical information from reliable sources such as WHO, NIH, NAMS, and peer-reviewed clinical guidelines using Retrieval-Augmented Generation (RAG).
- **Symptom Tracking & Personalized Feedback** – Allow users to log symptoms, mood, sleep quality, and activity through text or voice, providing real-time insights and personalized recommendations.
- **Preventive Health Support** – Offer guidance for maintaining bone health, cardiovascular health, hormonal balance, mental well-being, and preventive screenings while encouraging timely consultation with healthcare professionals when appropriate.
- **Multi-Agent System**
 - **Menopause Knowledge Agent** (retrieves evidence-based menopause information using RAG)
 - **Stage Assessment Agent** (determines the user's likely menopause stage)
 - **Wellness Recommendation Agent** (creates personalized diet, fitness, sleep, and lifestyle plans)
 - **Symptom Monitoring & Feedback Agent** (tracks symptoms, identifies trends, and provides adaptive recommendations)
- **Visualization Dashboard** – Display symptom trends, wellness scores, sleep quality, nutrition adherence, activity levels, and progress toward personal health goals through intuitive charts and reports.

Technology Used

Technology Used:

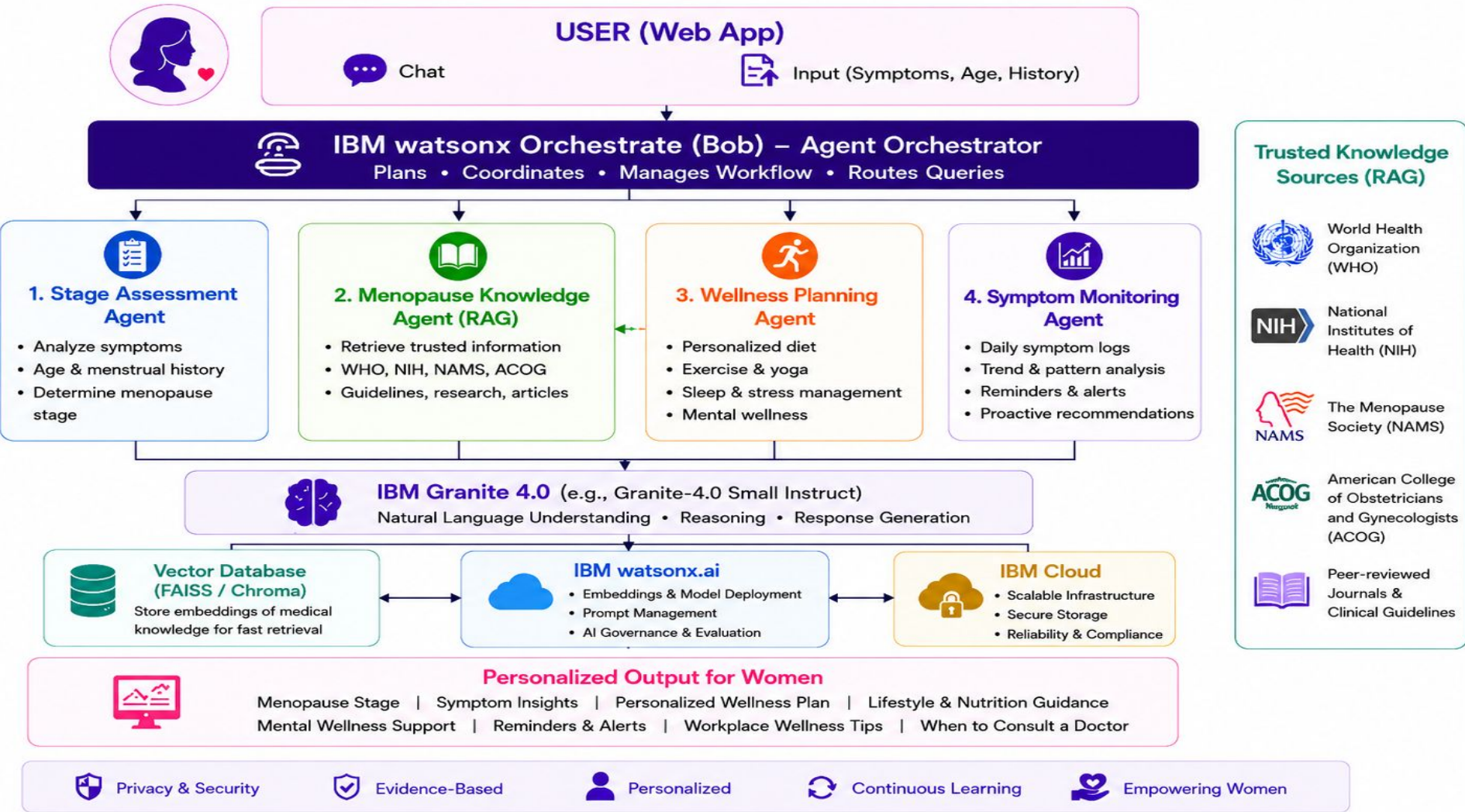
- **IBM watsonx studio, watson runtime and cloudobjectstorage:** For creating a new project on IBM Cloud
- **IBM watsonx Orchestrate (IBM Bob):** Multi-agent workflow design and orchestration.
- **IBM Cloud:** Secure, scalable infrastructure for deployment and data management.
- **IBM Granite model : Granite 4.0 h Small -** Natural language understanding, reasoning, and personalized recommendations.

Trusted Data Sources: WHO, NIH, The Menopause Society (NAMS), ACOG, and peer-reviewed clinical guidelines.

Architecture

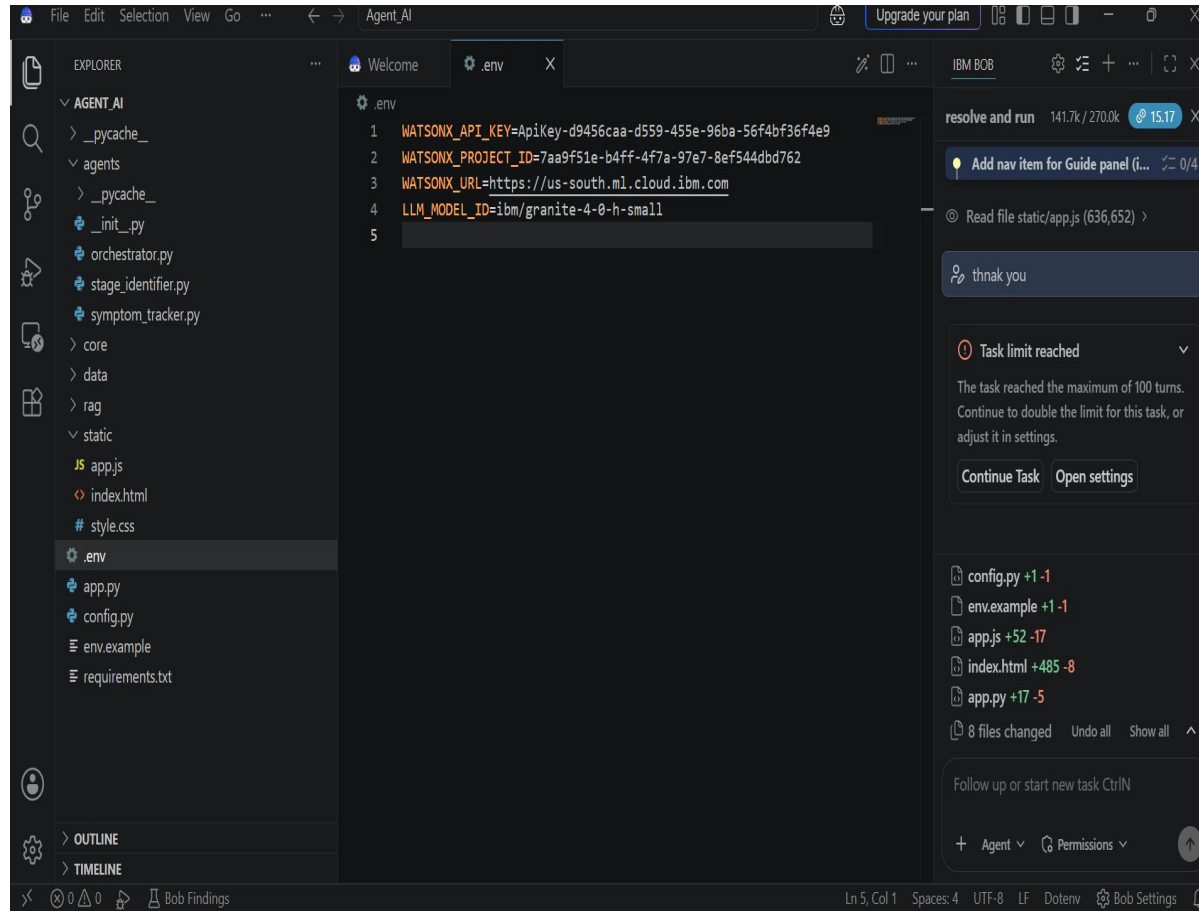
MenoStart – Agentic AI Architecture on IBM

IBM watsonx Orchestrate (Bob) Powered Multi-Agent System

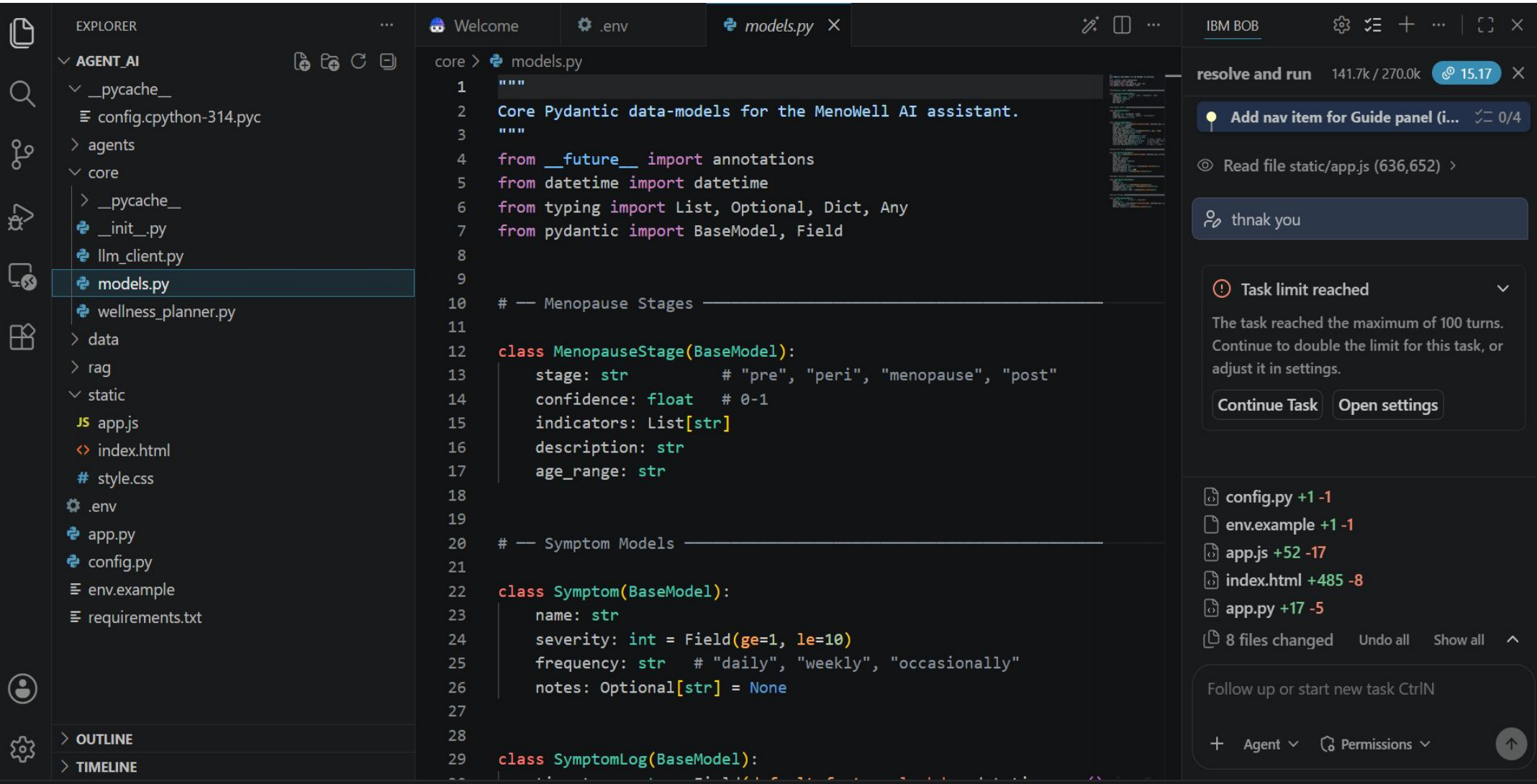


Required Files created using IBM bob -

1. App.py
2. Requirements.txt
3. Style.css
4. .env
5. Index.html



IBM Bob Screenshots:



EXPLORER

AGENT_AI

agents

__init__.py

orchestrator.py

stage_identifier.py

symptom_tracker.py

core

data

symptom_log.json

wellness_plans.json

rag

__pycache__

__init__.cpython-314.pyc

knowledge_base.cpython-314.pyc

__init__.py

knowledge_base.py

static

app.js

index.html

style.css

.env

app.py

config.py

env.example

requirements.txt

OUTLINE

TIMELINE

1

Welcome

.env

config.py

config.py

5nv

6

7

8

9nv("WATSONX_API_KEY", "ApiKey-d9456caa-d559-455e-96ba-56f4bf36f4e9")

10nv("WATSONX_PROJECT_ID", "7aa9f51e-b4ff-4f7a-97e7-8ef544dbd762")

11nv("WATSONX_URL", "https://us-south.ml.cloud.ibm.com")

12nv("LLM_MODEL_ID", "ibm/granite-4-0-h-small")

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27niLM-L6-v2"

28ma_db"

29/symptom_log.json"

30/wellness_plans.json"

31

IBM BOB

resolve and run141.7k / 270.0k15.17

Add nav item for Guide panel (i...0/4

Read file static/app.js (636,652)

thnak you

Task limit reached

The task reached the maximum of 100 turns. Continue to double the limit for this task, or adjust it in settings.

Continue TaskOpen settings

config.py +1 -1

env.example +1 -1

app.js +52 -17

index.html +485 -8

app.py +17 -5

8 files changedUndo allShow all

Restart IBM Bob to update!

Restart

Ln 26, Col 17Spaces: 4UTF-8IFEPythonRob Settings

User Interface for Menostart

 **MenoStart AI**

AI Assistant

Dashboard

Stage Assessment

Symptom Tracker

Wellness Plan

Menopause Guide

My Profile

Perimenopause

 Educational use only.

Always consult a healthcare professional.

AI Assistant

IBM

 Powered by WatsonX



Welcome to MenoStart AI

Your compassionate, evidence-based menopause wellness companion powered by IBM WatsonX Granite-8B.

Identify my stage

Nutrition advice

Sleep help

Mental wellbeing

Exercise plan

Health screenings

MenoStart Orchestrator

Ask me anything about menopause, symptoms, nutrition, exercise, or wellbeing...



Identifying stage though symptoms

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Stage Assessment

46

6

CYCLE REGULARITY

Irregular cycles

FSH LEVEL (IU/L)

Leave blank if unknown

Don't know your FSH? That's fine — leave it blank. FSH is measured via a blood test by your GP and helps confirm menopause stage (typical ranges: premenopause <10, perimenopause 10–40, postmenopause >40 IU/L).

CURRENT SYMPTOMS (SELECT ALL THAT APPLY)

☐ HOT FLASHES

☒ MOOD SWINGS

☒ ANXIETY

☐ WEIGHT GAIN

☒ HAIR THINNING

☐ NIGHT SWEATS

☒ BRAIN FOG

☒ VAGINAL DRYNESS

☒ HEART PALPITATIONS

☒ SKIN CHANGES

☒ IRREGULAR PERIODS

☒ MEMORY ISSUES

☒ REDUCED LIBIDO

☒ HEADACHES

☒ SLEEP PROBLEMS

☒ FATIGUE

☒ JOINT ACHES

☒ BLADDER ISSUES

Assess My Stage

IBM

Powered by WatsonX

Stage Assessment

**MenoStart AI**

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Stage Assessment

**Menopause Stage Assessment**
Answer a few questions to identify your menopause stage

Perimenopause

Confidence: 99%

✓ Irregular cycles and/or vasomotor symptoms present

✓ Symptom: Irregular Periods

✓ Symptom: Sleep Problems

✓ Symptom: Mood Swings

✓ Irregular periods

✓ Hot flashes / night sweats

Based on the symptoms described, this appears consistent with the **perimenopause** stage. Perimenopause typically begins in the mid-40s and is characterised by irregular periods, hot flashes, and sleep disturbances. I recommend tracking your cycle and consulting your GP for hormone level testing (FSH, LH, oestradiol).

Generate My Wellness Plan

Discuss with AI

Daily Symptom Tracker

 **MenoStart AI**

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Symptom Tracker

IBM

 Powered by WatsonX

Log Today's Symptoms

Hot Flashes

Severity: 310

Night Sweats

Severity: 110

Fatigue

Severity: 210

Mood Swings

Severity: 310

Brain Fog

Severity: 310

Joint Aches

Severity: 310

Sleep Issues

Severity: 310

Anxiety

Severity: 210

 MOOD SCORE (1-10)

7

 SLEEP HOURS

6h

NOTES

Any additional notes about today...

 **Medications & Health Context**

Helping the AI understand your full picture leads to better, safer recommendations.

Symptoms Trends

 **MenoStart AI**

AI Assistant

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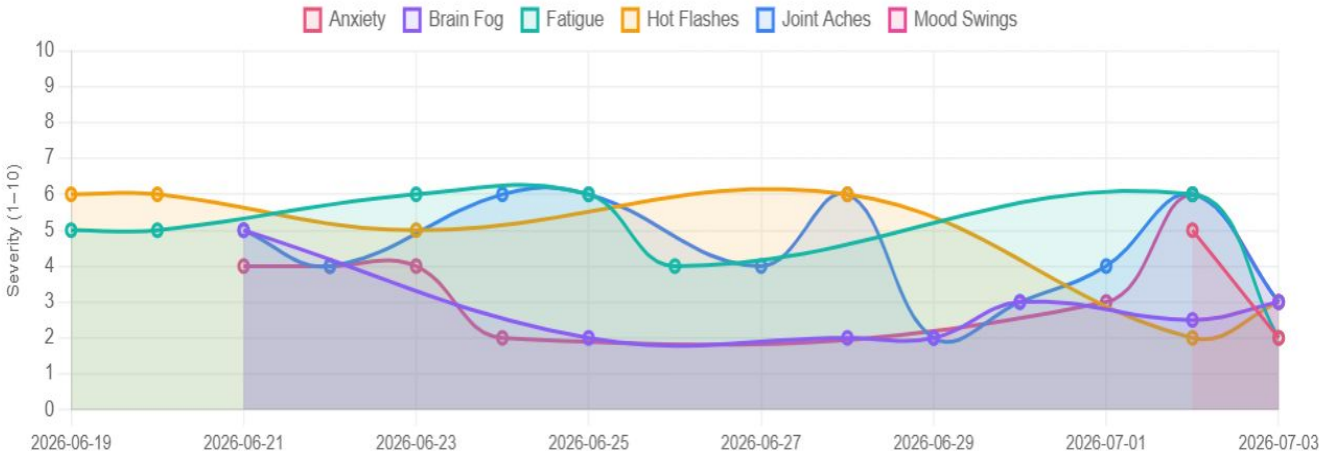
Always consult a healthcare professional.

Symptom Tracker

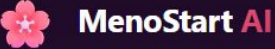
IBM Powered by WatsonX

Save Today's Log

Symptom Trends



Daily Wellness Plan - Nutrition , Exercise



AI Assistant

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Wellness Plan

 **Your Daily Wellness Plan**
Personalised for Perimenopause

 I embrace this transition with grace — my body is wise and resilient.

 **Nutrition**

- Increase calcium to 1200 mg/day — add yoghurt, kale, or fortified plant milk
- Eat soy-based foods (edamame, tofu) 3–4×/week for phytoestrogen support
- Include ground flaxseed (1–2 tbsp) daily on oats or salads
- Reduce refined sugar and processed foods to manage weight changes
- Eat oily fish (salmon, mackerel) twice a week for omega-3 benefits
- Ensure vitamin D 800–1000 IU daily — consider a supplement in winter

 **Exercise**

-  Low-impact exercise (swimming, cycling) to protect joints
- Strength training 3×/week — squats, lunges, rows for bone and muscle health
- 30-minute brisk walk daily — excellent for mood and cardiovascular health
- Yoga or Pilates 2×/week — reduces hot-flash frequency and improves sleep
- Pelvic floor exercises (Kegels) — 3 sets of 10, three times daily

Wellness Plan - Preventive Health

 **MenoStart AI**

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Wellness Plan

 **Preventive Health**

→ Discuss FSH/LH/oestradiol blood tests with your GP for confirmation

→ Ask about HRT options — review benefits and individual risks

→ Mammogram screening as per age guidelines

→ Annual cardiovascular risk check (BP, cholesterol, glucose)

→ DEXA bone density scan if you have risk factors

 **Evidence Sources**

 NICE Menopause Guideline NG23

 British Menopause Society 2024

 ACSM Exercise Guidelines

 National Osteoporosis Foundation

 CIPD Menopause at Work Guidelines 2023

 **Refresh Plan**

 Powered by WatsonX

Menopause Guide

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Menopause Guide

A complete awareness manual — understand every phase, symptoms, care, and when to seek help

 **Menopause Does Not Pause Your Life.** Understanding each phase empowers you to make informed decisions, seek timely care, and thrive through every stage of your journey.

Phase 1
Pre-Menopause Teens – Early 40s

▼

 Overview

Pre-menopause covers your entire reproductive years before the hormonal shifts of menopause begin. Oestrogen, progesterone, and FSH levels are in their normal cycling range. Periods are regular and predictable. This is the ideal time to build healthy habits that will protect your health through the transition ahead.

 Duration

From first menstrual period until perimenopause begins — typically **mid-40s**. There is no fixed end point; it is defined by the absence of menopausal symptoms.

 Common Symptoms

- Regular periods (21–35 day cycles)
- Normal energy and mood fluctuations tied to cycle
- Premenstrual symptoms (PMS) — bloating, breast tenderness

 Powered by WatsonX

Profile

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My Profile

IBM

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NAME

AGE

jayasree

46

CURRENT MENOPAUSE STAGE

Perimenopause

OTHER HEALTH CONDITIONS (select all that apply — your plan will be personalised accordingly)

☒  HYPERTENSION

☒  DIABETES / PRE-DIABETES

☐  THYROID DISORDER

☐  OSTEOPOROSIS

☐  HEART DISEASE

☐  OBESITY

☒  / WEIGHT CONCERN

☒  ARTHRITIS / JOINT PROBLEMS

☒  ANXIETY / DEPRESSION

STRESS RELIEF INTERESTS (select what you enjoy — we'll tailor stress management tips for you)

☒  MUSIC

☐  MEDITATION

☒  YOGA

☐  READING

☒  WALKING IN NATURE

☐  JOURNALING

☒  ART / CRAFT

☐  COOKING

☐  SOCIALISING

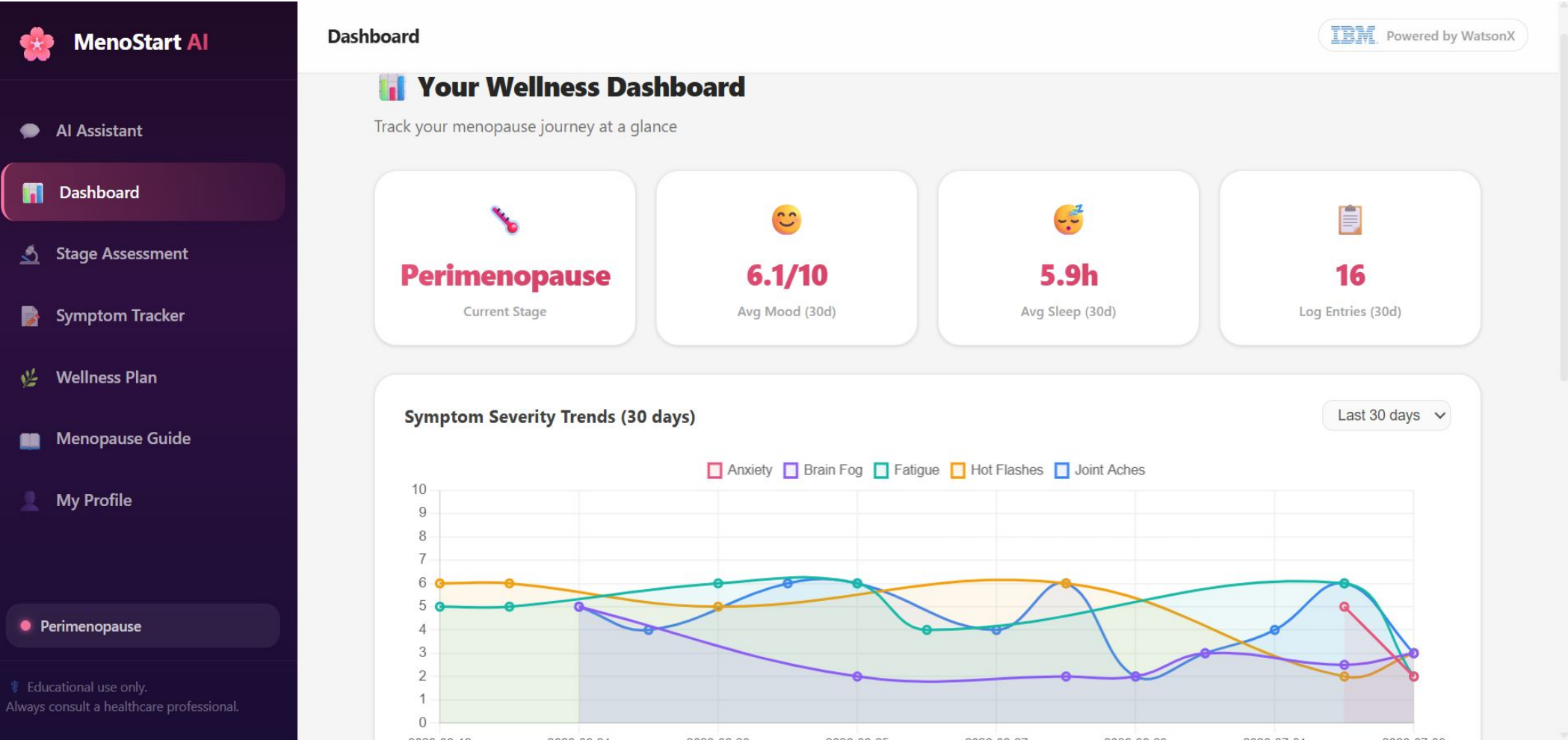
☐  GARDENING

☐  BREATHING EXERCISES

☐  DANCING

Save Profile

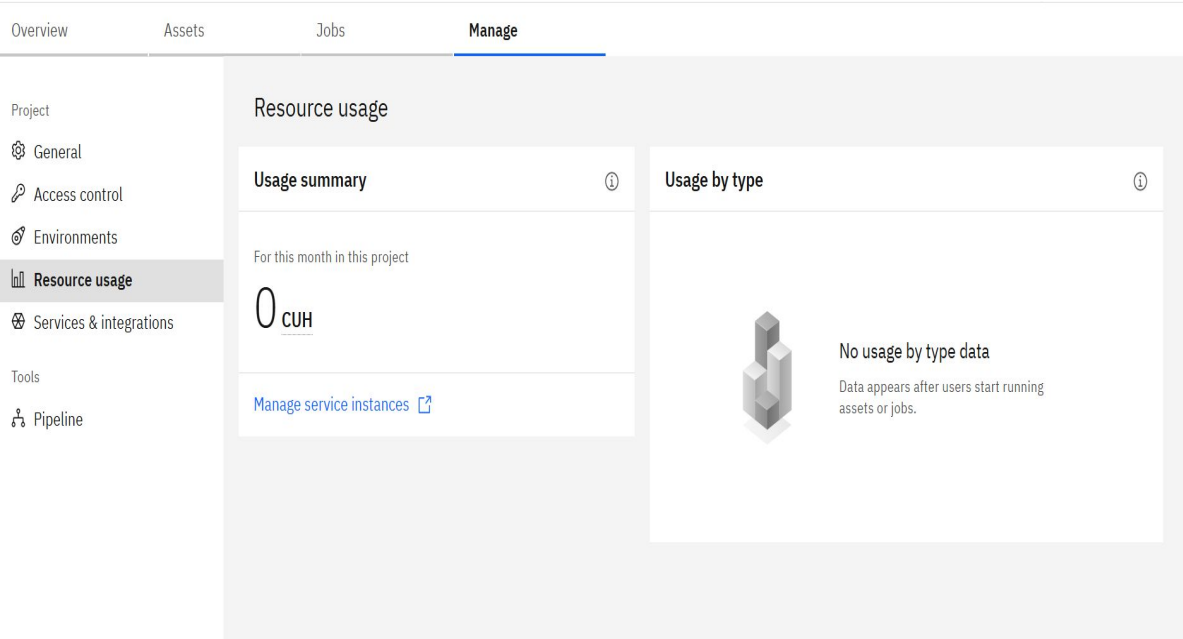
Personalized Wellness Dashboard



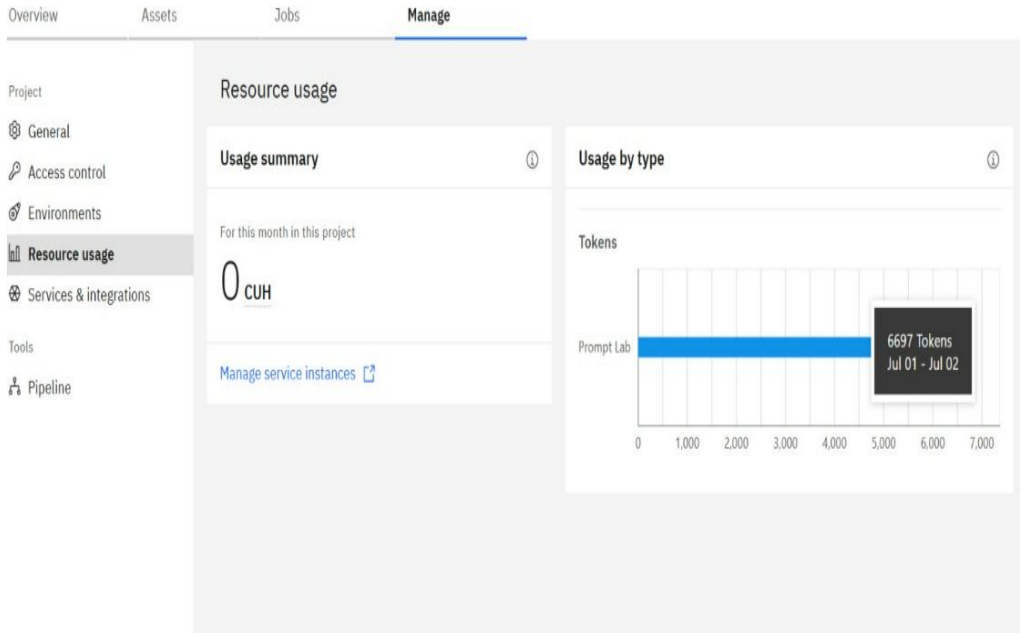
Token Usage

6697 token are used

Before Prompt



After the Prompt



Role of Agentic AI in the solution

Agentic AI serves as the intelligent core of **MenoStart**, enabling multiple specialized AI agents to collaborate in providing personalized, evidence-based menopause care. It assesses the user's menopause stage based on symptoms and health information, retrieves trusted medical knowledge using RAG, generates adaptive wellness plans for nutrition, exercise, sleep, and mental well-being, continuously tracks symptom progression, and proactively offers health insights, reminders, and workplace wellness recommendations. By learning from user interactions and coordinating these tasks autonomously, Agentic AI delivers continuous, explainable, and personalized support throughout every stage of the menopause journey.



Novelty and Uniqueness

MenoStart goes beyond a traditional symptom checker or chatbot by leveraging **Agentic AI** to deliver continuous, personalized, and evidence-based menopause support. It combines **multi-agent collaboration**, **RAG-powered retrieval of trusted medical knowledge**, and **adaptive wellness planning** to understand each user's unique menopause journey. Unlike generic health apps, MenoStart identifies the user's menopause stage, tracks symptom progression over time, generates personalized diet, exercise, sleep, and mental wellness plans, and proactively provides preventive health and workplace wellness recommendations. Its ability to learn from user interactions, explain its recommendations, and retrieve information only from reliable medical sources makes it a trustworthy, intelligent, and holistic digital companion for women throughout the menopause transition.

Git Hub Link

<https://github.com/Jayasree1811/Menostart.git>

Contents:

1. Generative AI in Action Credly Certificate
2. IBM BOB Trouble Shoot Certificate
3. Project PPT
4. Problem Statement

Future Scope

1.Enhanced Accessibility: Integrate **multilingual support** and a **voice-enabled interface (speech-to-text and text-to-speech)** to make MenoStart accessible to women from diverse linguistic backgrounds, including those with limited digital literacy or visual impairments.

2.Advanced Personalized Healthcare: Integrate **wearable devices, telemedicine,** and **predictive AI** to enable real-time health monitoring, early risk detection, personalized wellness recommendations, and seamless consultation with healthcare professionals.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Jayasree H

Has successfully satisfied the requirements for:

Generative AI in Action



Issued on: Jun 30, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/c5640c62-2269-48bf-bef0-ade68a5c9113>



Certificates Earned:

IBM SkillsBuild	Completion Certificate
	This certificate is presented to Jayasree Hanumantha Rao for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 01 Jul 2026 (GMT)	Learning hours: 30 mins



This certificate is presented to
Jayasree Hanumantha Rao

for the completion of
**Getting Started with Cybersecurity (Earn a
credential!)**

(PLAN-C7EE7CC95370)

According to the Your Learning Builder - Plans system of record

Completion date: 01 Jul 2026 (GMT)



This certificate is presented to

Jayasree Hanumantha Rao

for the completion of

Level Up Cybersecurity with Generative AI

(ALM-COURSE_3947352)

According to the Adobe Learning Manager system of record

Completion date: 04 Jul 2026 (GMT)

Learning hours: 1 hr 30 mins



This certificate is presented to
Jayasree Hanumantha Rao

for the completion of
Introduction to Artificial Intelligence
(ALM-COURSE_4058918)

According to the Adobe Learning Manager system of record



This certificate is presented to
Jayasree Hanumantha Rao

for the completion of
Getting Started with Artificial Intelligence

(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of record

IBM SKILLS BUILD PROFILE

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to
Jayasree Hanumantha Rao

for the completion of
Exploring Quantum Computing
(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 02 Jul 2026 (GMT)

Learning hours: 50 mins

IBM SKILLS BUILD PROFILE

IBM **SkillsBuild**

Profile



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Telangana, India

ACA - Edunet Foundation

Thank You!

Thank you for your time and interest.